Induction Heads as an Essential Mechanism for Pattern Matching in In-context Learning

Joy Crosbie[⋆]
ILLC
University of Amsterdam
joy.m.crosbie@gmail.com

Ekaterina Shutova
ILLC
University of Amsterdam
e.shutova@uva.nl

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Abstract

Large language models (LLMs) have shown a remarkable ability to learn and perform complex tasks through in-context learning (ICL). However, a comprehensive understanding of its internal mechanisms is still lacking. This paper explores the role of *induction heads* in a few-shot ICL setting. We analyse two state-of-the-art models, Llama-3-8B and InternLM2-20B on abstract pattern recognition and NLP tasks. Our results show that even a minimal ablation of induction heads leads to ICL performance decreases of up to ~32% for abstract pattern recognition tasks, bringing the performance close to random. For NLP tasks, this ablation substantially decreases the model's ability to benefit from examples, bringing few-shot ICL performance close to that of zero-shot prompts. We further use *attention knockout* to disable specific induction patterns, and present fine-grained evidence for the role that the induction mechanism plays in ICL.

1 Introduction

Large language models have shown a remarkable ability to learn and perform complex tasks through in-context learning (ICL) (Brown et al., 2020; Touvron et al., 2023b). In ICL, the model receives a demonstration context and a query question as a prompt for prediction. Unlike supervised learning, ICL utilises the pretrained model's capabilities to recognise and replicate patterns within the demonstration context, thereby enabling accurate predictions for the query without the use of gradient updates.

Given the success and wide applicability of ICL, much recent research was dedicated to better understanding its properties. Several works explored its connections to gradient descent (Dai et al., 2023; Von Oswald et al., 2023), suggesting that ICL functions as an implicit form of fine-tuning at inference time. Other works investigated factors influencing ICL, showing that it is driven by the distributions of the training data (Chan et al., 2022) and scales with model size, revealing new abilities at certain parameter thresholds (Brown et al., 2020; Wei et al., 2022). During inference, the properties of demonstration samples also affect ICL performance , with aspects such as the label space, input distribution and input-label pairing playing a crucial role (Min et al., 2022; Webson and Pavlick, 2022). While this work identified interesting properties of ICL and effective ICL prompting strategies, a comprehensive understanding of its operational mechanisms within the models is still lacking.

Our paper aims to fill this gap, by directly investigating the internal model computations that enable ICL. Our work is inspired by recent research in the field of mechanistic interpretability, which aims to reverse engineer the "algorithm" by which Transformer models process information (Geva et al., 2023; Olah et al., 2020; Wang et al., 2022). As a significant milestone in this area, Elhage et al. (2021) demonstrated the existence of *induction heads* in Transformer LMs. These heads scan the context for previous instances of the current token using a *prefix matching* mechanism, which identifies if and where a token has appeared before. If a matching token is found, the head employs a *copying* mechanism to increase the probability of the subsequent token, facilitating exact or approximate repetition of sequences and embodying the algorithm "[A][B]...[A] → [B]".

Building on this foundation, Olsson et al. (2022) hypothesised that induction heads are capable of abstract pattern matching and conducted a qualitative analysis of attention patterns observed in an example from an abstract classification task. They observed that these heads, when focused on the final token, tended to attend to previous instances

[⋆] Corresponding author.
Code: github.com/JoyC177/Induction_Heads_ICL

of the correct label. However, they did not conduct any experiments to verify this. In a related study, Bansal et al. (2023) investigated which heads were important for NLP tasks and found some degree of overlap between these heads and those that had high scores associated with induction. However, they did not directly assess the contribution of induction heads to ICL performance of the models in these tasks.

To the best of our knowledge, our paper is the first one to directly, empirically link the specific computations performed by induction heads to measurable improvements in ICL performance. We investigate the role of induction heads as an underlying mechanism of few-shot ICL across a range of tasks: (1) abstract pattern recognition tasks; (2) popular NLP tasks. We focus on two state-of-the-art models: Llama-3-8B (Dubey et al., 2024) and InternLM2-20B (Cai et al., 2024). We first compute prefix matching scores for all of their attention heads, and then perform head ablation experiments by removing 1% and 3% of the heads with the highest scores. For the NLP tasks, we additionally conduct experiments with semantically unrelated labels to encourage the models to rely on ICL for task completion. To confirm our hypothesis regarding the specific role of the induction mechanisms in these heads, we further perform *attention knockout* experiments, where we selectively inhibit each token's ability to attend to any tokens that directly followed tokens similar to the current token. This effectively simulates a loss of function in the prefix matching and copying mechanisms.

We find that the ablation of induction heads results in a substantial decrease in ICL performance, much more so than when an equivalent percentage of random heads are ablated. Additionally, when we block the induction attention pattern in these heads, performance drops to levels comparable to or worse than those seen with full head ablations. The latter confirms not merely the importance of induction heads for ICL, but also the specific role of the prefix matching and copying mechanism, retracing its application at the token level. Our work is thus the first to provide comprehensive experimental evidence, demonstrating (1) the essential role of induction heads in few-shot ICL in large-scale, widely-used LMs and real-world tasks and (2) that induction heads utilise a "fuzzy" version of the prefix matching and copying mechanisms to enable pattern matching.

2 Related Work

2.1 Factors that influence ICL performance

ICL performance is influenced by various factors, both during the pretraining and inference stage. During pretraining, ICL abilities emerge when training data displays specific distributional properties, such as items frequently appearing in clusters and the presence of many infrequently occurring classes (Chan et al., 2022). Additionally, the scalability of these abilities correlates directly with model size (Brown et al., 2020), and emergent capabilities become apparent at particular scales of pretraining or beyond certain parameter thresholds (Wei et al., 2022; Lu et al., 2023). In the inference stage, the characteristics of demonstration examples, such as label space, input distribution, and input-label pairing format, significantly impact ICL performance (Min et al., 2022). Additionally, models are more affected by label choice than by instruction semantics (Webson and Pavlick, 2022), and larger models can adapt to input-label mappings even with flipped or unrelated labels (Wei et al., 2023b). Further, fine-tuning on semantically unrelated in-context input-label pairs enhances performance on new ICL tasks (Wei et al., 2023a).

2.2 Mechanistic Interpretability

Research in mechanistic interpretability seeks to explain model behaviours in terms of their internal components (Olah et al., 2020). Elhage et al. (2021) demonstrated that small, attention-only Transformers can be deconstructed into *circuits*—specific sub-graphs within the model's computational graph that perform distinct tasks (Wang et al., 2022). By analysing these circuits, Elhage et al. demonstrated the existence of induction heads. Circuit discovery has since led to the localisation of other behaviours within Transformer models (Meng et al., 2022; Vig et al., 2020; Wang et al., 2022). A common technique is *mean ablation*, where components' activations are replaced with their average activation value across a reference distribution to determine whether disrupting this specific component hinders the model's ability to perform the task. Wang et al. (2022) studied indirect object identification in GPT-2 small (Radford et al., 2019), by obscuring the indirect object with random names, averaging these activations, and replacing the target activations during inference. This identified specific attention heads crucial for the task.

Another recent technique—*attention knockout*—

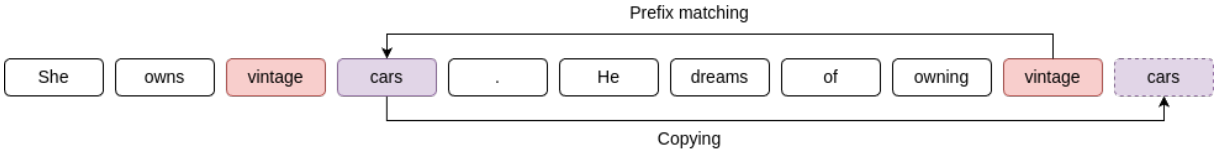

Figure 1: In the sequence "...vintage cars ... vintage", an induction head identifies the initial occurrence of "vintage", attends to the subsequent word "cars" for prefix matching, and predicts "cars" as the next word through the copying mechanism.

systematically disables specific attention weights to assess their impact on model outputs (Geva et al., 2023). Geva et al. (2023) explored how factual knowledge is extracted from LLMs during inference, by blocking the final position from attending to both subject and non-subject positions across specific layers. They revealed that essential information from non-subject positions is accessed first. Additionally, Wang et al. (2023) blocked label tokens from accessing prior demonstration text in shallow layers, and demonstrated that label words gather information early in forward propagation.

2.3 Induction heads & ICL

Building on the work of Elhage et al. (2021), Olsson et al. (2022) provided initial evidence that induction heads might be an underlying mechanism of ICL, defined as the gradual reduction of loss with increasing token indices. They observed that induction heads emerge early in training, coinciding with significant improvements in the model's ability to learn from context. They also found that modifying or removing these heads impairs ICL performance, assessed by a heuristic measure of loss reduction. In contrast, we employ a few-shot prompting ICL setting and evaluate ICL performance directly through the model's accuracy on specific tasks.

Using head importance pruning, Bansal et al. (2023) demonstrated that up to 70% of attention heads and 20% of feed forward layers can be removed from the OPT-66B model (Zhang et al., 2022) with minimal performance decline across various downstream tasks. They uncovered a notable consistency in the relevance of certain attention heads for ICL, irrespective of the task or the number of few-shot examples. They investigated whether these crucial heads were induction heads, finding some degree of overlap with those that had high prefix matching scores. Our research extends these findings by not only identifying induction heads but also conducting ablation studies to directly evaluate their impact on ICL performance in diverse tasks.

3 Background

Following the framework established by Elhage et al. (2021) and further discussed by Ren et al. (2024), the operations within a multi-head attention (MHA) layer, comprising H attention heads, can be reformulated as follows:

$$\sum_{h=1}^{H} \text{softmax}\left(\frac{\mathbf{x}W_q^h(\mathbf{x}W_k^h)^T}{\sqrt{d_h}} + M^h\right)\mathbf{x}W_v^hW_o^h$$
$$= \sum_{h=1}^{H} \text{softmax}\left(\frac{\mathbf{x}W_{QK}^h\mathbf{x}^T}{\sqrt{d_h}} + M^h\right)\mathbf{x}W_{OV}^h \tag{1}$$

where $\mathbf{x} = [\mathbf{x}_1^T, \mathbf{x}_2^T, ..., \mathbf{x}_N^T]^T \in \mathbb{R}^{N\times d}$ represents the sequence of embeddings, with each $\mathbf{x}_i = \mathbf{t}_i\mathbf{W}_e \in \mathbb{R}^{1\times d}$ denoting the embedding of the i-th input token t_i. $\mathbf{W}_e \in \mathbb{R}^{|\mathcal{V}|\times d}$ denotes the embedding matrix over the vocabulary $\mathcal{V}$. $\mathbf{W}_q^h, \mathbf{W}_k^h, \mathbf{W}_v^h \in \mathbb{R}^{d\times d_h}$ denote the query, key, and value parameter matrices of the h-th attention head. $\mathbf{W}_o \in \mathbb{R}^{d\times d}$ represents the output transformation of the MHA layer, which can be deconstructed as $\mathbf{W}_o = [(\mathbf{W}_o^1)^T(\mathbf{W}_o^2)^T...(\mathbf{W}_o^H)^T]$, where $\mathbf{W}_o^h \in \mathbb{R}^{d_h\times d}$. M^h is a casual attention mask, which ensures each position can only attend to preceding positions (i.e. $M_{rc}^h = -\infty\, \forall c > r$ and zero otherwise).

In Equation (1), $\mathbf{W}_{QK}^h = \mathbf{W}_q^h(\mathbf{W}_k^h)^T$, termed the Query-Key (QK) circuit, calculates the attention pattern of the h-th head. The matrix $\mathbf{W}_{OV}^h = \mathbf{W}_v^h(\mathbf{W}_o^h)^T$, termed the Output-Value (OV) circuit, determines each head's independent output for the current token. Leveraging this decomposition, Elhage et al. (2021) discovered a distinct behaviour in certain attention heads, which they named *induction heads*. This behaviour emerges when these heads process sequences of the form "[A] [B] ... [A] $\rightarrow$ ". In these heads, the QK circuit directs attention towards [B], which appears directly after